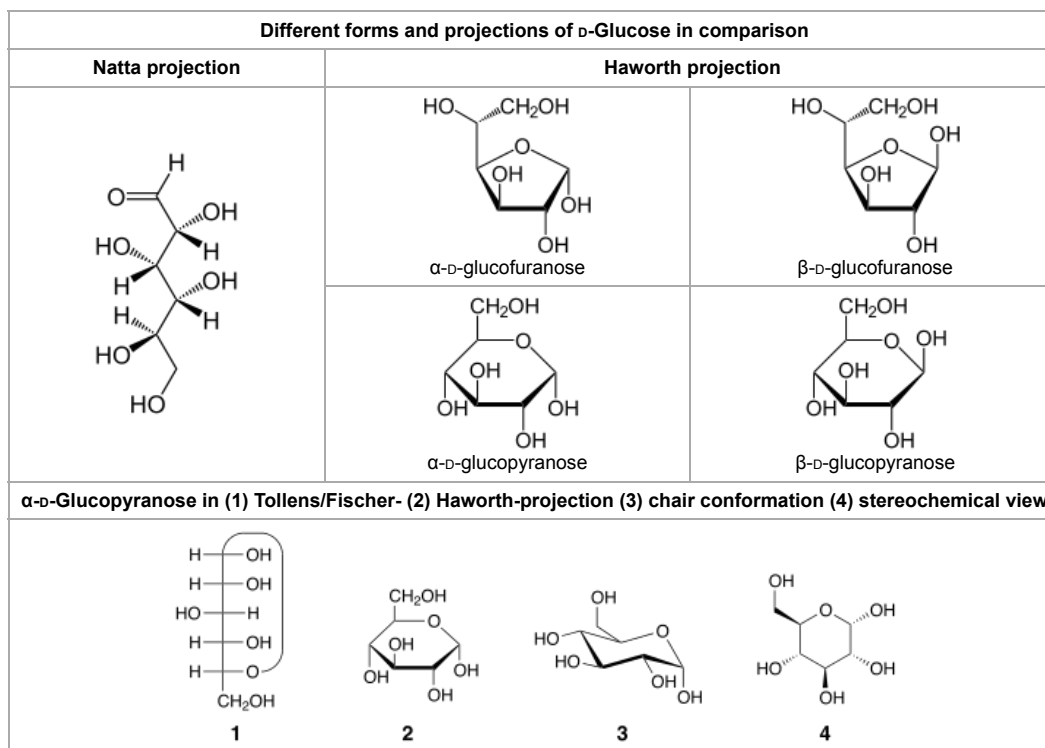




Open-chain form

In its fleeting open-chain form, the glucose molecule has an open (as opposed to cyclic) and unbranched backbone of six carbon atoms, C-1 through C-6; where C-1 is part of an aldehyde group $\text{H}(\text{C}=\text{O})$ -, and each of the other five carbons bears one hydroxyl group -OH. The remaining bonds of the backbone carbons are satisfied by hydrogen atoms -H. Therefore, glucose is both a hexose and an aldose, or an aldohexose. The aldehyde group makes glucose a reducing sugar giving a positive reaction with the Fehling test.

Each of the four carbons C-2 through C-5 is a stereocenter, meaning that its four bonds connect to four different substituents. (Carbon C-2, for example, connects to $-(\text{C}=\text{O})\text{H}$ -, -OH-, -H-, and $-(\text{CHOH})_4\text{H}$.) In D-glucose, these four parts must be in a specific three-dimensional arrangement. Namely, when the molecule is drawn in the Fischer projection, the hydroxyls on C-2, C-4, and C-5 must be on the right side, while that on C-3 must be on the left side.

The positions of those four hydroxyls are exactly reversed in the Fischer diagram of L-glucose. D- and L-glucose are two of the 16 possible aldohexoses; the other 14 are allose, altrose, galactose, gulose, idose, mannose, and talose, each with two enantiomers, "D-" and "L-".

It is important to note that the linear form of glucose makes up less than 0.02% of the glucose molecules in a water solution. The rest is one of two cyclic forms of glucose that are formed when the hydroxyl group on carbon 5 (C5) bonds to the aldehyde carbon 1 (C1).

Cyclic forms

In solutions, the open-chain form of glucose (either "D-" or "L-") exists in equilibrium with several cyclic isomers, each containing a ring of carbons closed by one oxygen atom. In aqueous solution however, more than 99% of glucose molecules, at any given time, exist as pyranose forms. The open-chain form is limited to about 0.25% and furanose forms exists in negligible amounts. The terms "glucose" and "D-glucose" are generally used for these cyclic forms as well. The ring arises from the open-chain form by an intramolecular nucleophilic addition reaction between the aldehyde group (at C-1) and either the C-4 or C-5 hydroxyl group, forming a hemiacetal linkage, $-\text{C}(\text{OH})\text{H}-\text{O}-$.

The reaction between C-1 and C-5 yields a six-membered heterocyclic system called a pyranose, which is a monosaccharide sugar (hence "-ose") containing a derivatised pyran skeleton. The (much rarer) reaction between C-1 and C-4 yields a five-membered furanose ring, named after the cyclic ether furan. In either case, each carbon in the ring has one hydrogen and one hydroxyl attached, except for the last carbon (C-4 or C-5) where the hydroxyl is replaced by the remainder of the open molecule (which is $-(\text{C}(\text{CH}_2\text{OH})\text{HOH})-\text{H}$ or $-(\text{CHOH})-\text{H}$, respectively).

The ring-closing reaction makes carbon C-1 chiral, too, since its four bonds lead to -H, to -OH, to carbon C-2, and to the ring oxygen. These four parts of the molecule may be arranged around C-1 (the anomeric carbon) in two distinct ways, designated by the prefixes "α-" and "β-". When a glucopyranose molecule is drawn in the Haworth projection, the designation "α-" means that the hydroxyl group attached to C-1 and the $-\text{CH}_2\text{OH}$ group at C-5 lies on opposite sides of the ring's plane (a *trans*

formula	
Molar mass	180.156 g·mol ^{−1}
Appearance	White powder
Density	1.54 g/cm ³
Melting point	α-D-glucose: 146 °C (295 °F; 419 K) β-D-glucose: 150 °C (302 °F; 423 K)
Solubility in water	909 g/L (25 °C (77 °F))
Magnetic susceptibility (χ)	−101.5×10 ^{−6} cm ³ /mol
Dipole moment	8.6827
Thermochemistry	
Heat capacity (C)	218.6 J K ^{−1} mol ^{−1} [1]
Std molar entropy (S° ₂₉₈)	209.2 J K ^{−1} mol ^{−1} [1]
Std enthalpy of formation (Δ _f H° ₂₉₈)	−1271 kJ/mol[2]
Heat of combustion, higher value (HHV)	2,805 kJ/mol (670 kcal/mol)
Pharmacology	
ATC code	B05CX01 (WHO (https://www.whocc.no/atc_ddd_index/?code=B05CX01)) V04CA02 (WHO (https://www.whocc.no/atc_ddd_index/?code=V04CA02)), V06DC01 (WHO (https://www.whocc.no/atc_ddd_index/?code=V06DC01))
Hazards	
Safety data sheet	ICSC 08655 (http://www.inchem.org/documents/icsc/icsc/eics0865.htm)
NFPA 704 (fire diamond)	
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).	
✓ verify (what is ✗?) Infobox references	